

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 09—REAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B) and (C) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 09—REAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Let A be the open interval $(-1, 1)$ then, the cluster points of A are :
 - (A) 1 and -1 only.
 - (B) All the points of $(-1, 1)$.
 - (C) All the points of $[-1, 1]$.
2. Which of the following have no cluster points ?
 - (A) $(3, 4]$.
 - (B) $[2, 3]$.
 - (C) $\{2, 3, 4\}$.
3. Let $f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then $\lim_{x \rightarrow 0} f(x)$ is :
 - (A) 1.
 - (B) -1 .
 - (C) None of these.
4. Given a, b satisfy $0 < a < 1, b > 1$. Then which of the following sequences is convergent ?
 - (A) $\frac{ab^n}{2^n}$.
 - (B) $\frac{2^{3n}}{3^{2n}}$.
 - (C) $\frac{n}{b^n}$.
5. Let $A \subseteq \mathbb{R}$ and $C \in \mathbb{R}$ such that for every $\delta > 0$ there exist atleast one point $x \in A, x \neq c$ such that $|x - c| < \delta$. Then c is a _____ of A .
 - (A) Isolated point.
 - (B) Cluster point.
 - (C) None of these.
6. When the graph of a function f in $\mathbb{R}$ has a break such that the function value and the limit (does not exist) aren't the same, then f is said to have a _____ discontinuity at that point.
 - (A) Removable.
 - (B) Jump.
 - (C) None of these.
7. Consider $f(x) = 6x^3 - 3x^2 + 2$. Then which of the following is true ?
 - (A) f has a zero in $(0, 1)$.
 - (B) f has no zero.
 - (C) f has a zero in $(-1, 1)$.

8. Which of the following statements is true ?

- (A) Every continuous function is uniformly continuous.
- (B) Every uniformly continuous function is continuous.
- (C) None of these.

9. Let $f : [a, b] \rightarrow \mathbb{R}$. If there exist a $L \in \mathbb{R}$ such that for every $\epsilon > 0$, there exist $\delta_\epsilon > 0$ such that for any tagged partition $\dot{p}$ of $[a, b]$ with $\|\dot{p}\| < \delta_\epsilon$; $|s(f; p) - L| < \epsilon$, then ' f ' is :

- (A) Riemann Integrable.
- (B) Lebesgue Integrable.
- (C) None of these.

10. Let $g : [0, 4] \rightarrow \mathbb{R}$ be defined by $g(x) = 3$ for $0 \leq x \leq 2$ and $g(x) = 2$ for $2 < x \leq 4$ then $\int_0^4 g =$

- (A) 5.
- (B) 7.
- (C) 10.

11. Which of the following is true ?

- (A) $f \in R[a, b] \Rightarrow f$ is bounded.
- (B) $f \in R[a, b] \Rightarrow f$ is continuous.
- (C) $f \in R[a, b] \Rightarrow f$ is monotone.

12. Let $x \in [0, 1]$ and ' f ' be such that $f(x) = \begin{cases} 1, & x \text{ is rational} \\ 0, & x \text{ is irrational} \end{cases}$. Then :

- (A) $f \in R[0, 1]$.
- (B) $f \notin R[0, 1]$.
- (C) None of these.

13. Suppose $f : [a, b] \rightarrow \mathbb{R}$ and $f(x) = 0$ except for finite number of points $c_1, c_2, c_3, \dots, c_n$ in $[a, b]$. Then :

- (A) $f \notin R[a, b]$.
- (B) $\int_a^b f = 0$.
- (C) None of these.

14. Which of the following is true ?

- (A) Every convergent sequence is Cauchy.
- (B) Every Cauchy sequence is convergent.
- (C) Both (A) and (B).

Turn over

15. Let $f : [a, b] \rightarrow \mathbb{R}$. Then which of the following is true ?

- (A) f is continuous on $[a, b] \Rightarrow f \in R[a, b]$.
- (B) $f \in R[a, b] \Rightarrow f$ is continuous on $[a, b]$.
- (C) $f \in R[a, b] \Rightarrow f$ is uniformly continuous on $[a, b]$.

16. Consider 'h' defined on $[0, 1]$ by $h(x) = \begin{cases} x+1, & x \text{ is rational} \\ 0, & x \text{ is irrational.} \end{cases}$ Then which of these is true ?

- (A) $h \in R[0, 1]$.
- (B) $h \notin R[0, 1]$.
- (C) h is continuous on $[0, 1]$.

17. Let H be defined on $[0, 1]$ such that $H(x) = \begin{cases} k, & \text{for } x = \frac{1}{k}, k \in \mathbb{R} \\ 0, & \text{elsewhere} \end{cases}$. Then which of these is true ?

- (A) $H \in R[0, 1]$.
- (B) $H \notin R[0, 1]$.
- (C) H continuous on $[0, 1]$.

18. If f is bounded on $[a, b]$ and if ' f ' restricted to $[c, b]$, $c \in [a, b]$ is Riemann integrable, then which of the following does not hold true ?

- (A) $f \in R[a, b]$.
- (B) $f \notin R[a, b]$.
- (C) $\lim_{c \rightarrow a} \int_c^b f = \int_a^b f$.

19. If ' f ' is continuous of every point of $[a, b]$ and F is any antiderivative of ' f ' on $[a, b]$, then $\int_a^b f(x) \cdot dx =$

- (A) $(b - a)(F(b) - F(a))$.
- (B) $F(b) - F(a)$.
- (C) None of these.

20. If $g(x) = \begin{cases} x, & |x| \geq 1 \\ -x, & |x| < 1 \end{cases}$ and if $G(x) = \frac{1}{2}|x^2 - 1|$, then $\int_{-2}^3 g(x) \cdot dx =$

- (A) $\frac{1}{2}$.
- (B) $\frac{5}{2}$.
- (C) $\frac{7}{2}$.

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 09—REAL ANALYSIS

Time : Three Hours

Maximum : 120 Marks

Section A

Answer all questions.

Each question carries 1 mark.

1. State Bolzano's Intermediate Value Theorem.
2. Define Uniformly continuous function.
3. Define Step function on an interval.
4. State the Composition theorem of Riemann integration.
5. Define Partition of a closed and bounded interval.
6. Define Indefinite integral of $f \in R[a, b]$, with base point a .
7. State the Squeeze theorem for Riemann integrability of a function.
8. State the Cauchy criterion for Riemann integrability of a function.
9. For what value of r the series $\sum_{n=1}^{\infty} r^n \cos^2 nx$ uniformly convergent ?
10. Fill in the blanks : $\lim \left(\frac{\cos (nx + n)}{n} \right) =$ _____.
11. Fill in the blanks : $\beta (1, 1) =$ _____.
12. Fill in the blanks : $\int_{-\infty}^{\infty} e^{-x^2} dx =$ _____.

(12 × 1 = 12 marks)

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Section B

*Answer at least **eight** questions.*

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 48.

13. State the Boundedness Theorem. Show by an example that the boundedness theorem fails if the function is discontinuous in that interval.
14. State the Maximum Minimum Value Theorem. Show by an example that this theorem fails, if the interval is not closed.
15. Prove that the continuous image of a closed and bounded interval is a closed and bounded interval.
16. State the “non-uniform continuity criteria”. Apply this result to test the uniform continuity of $f(x) = 1/x$ on $(0, \infty)$.
17. State the location of Roots Theorem. Use it to show that the equation $xe^x - 2 = 0$ has a root in $(0, 1)$.
18. If $f, g \in R[a, b]$, then prove that $f \cdot g \in R[a, b]$.
19. Evaluate $\int_{t=1}^4 \frac{\sin \sqrt{t}}{\sqrt{t}} dt$.
20. If $f \in R[a, b]$, then prove that $|f| \in R[a, b]$.
21. Define Pointwise convergence of a sequence of functions. Test the convergence of (x^n) .
22. If (f_n) and (g_n) are uniformly continuous functions in R , then show that $(f_n)(g_n)$ need not be uniformly continuous in R .
23. Distinguish between the absolute and uniform convergence of a series of functions ; also state the Cauchy criteria for uniform convergence of a series of functions.
24. Show that $\int_{x=a}^b \frac{1}{\sqrt{(b-x)^p}} dx$ converges, if $p < 1$; and diverges if $p \geq 1$.

25. Express the integral $\int_{x=1}^2 \frac{x^2}{(1-x^5)} dx$, in terms of Beta function.

26. Define Gamma function. Prove that $\Gamma(n) = (n-1) \Gamma(n-1)$.

(8 × 6 = 48 marks)

Section C

Answer at least five questions.

Each question carries 9 marks.

All questions can be attended.

Overall Ceiling 45.

27. State and prove the uniform continuity theorem.

28. (a) Define Lipschitz function.

(b) If $f : A \rightarrow \mathbb{R}$ Lipschitz function on A , then prove that f is uniformly continuous on A .

29. If $f, g \in \mathbb{R}[a, b]$, then prove that $f + g \in \mathbb{R}[a, b]$ and $\int_a^b f + g = \int_a^b f + \int_a^b g$.

30. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then prove that $f \in \mathbb{R}[a, b]$.

31. Test the uniform convergence of the series $\sum_{n=1}^{\infty} \frac{\sin(x^2 + nx)}{n(n+2)}$.

32. Discuss the convergence of $(f_n(x)) = (x^n(1-x))$, $x \in A = [0, 1]$.

33. (a) Distinguish between absolute and conditional convergence of an improper integral.

(b) Discuss the convergence of the Improper integral $\int_{x=\pi}^{\infty} \frac{\cos x}{x^2} dx$.

Turn over

34. Express the integral $\int_0^a x^{p-1} (a-x)^{q-1} dx$; $p, q > 0$ in terms of Gamma function.

35. Prove that $\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$.

(5 × 9 = 45 marks)

Section D

Answer any **one** question.

The question carries 15 marks.

36. (a) If $f: A \rightarrow \mathbb{R}$ is uniformly continuous on A and if (x_n) is a Cauchy sequence in A , then prove that $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.

(b) Evaluate $\int_0^\infty e^{-x^2} dx$, Gamma function.

37. (a) State and prove the Fundamental Theorem of calculus (2nd form).

(b) Discuss the uniform convergence $(f_n(x)) = (x^n)$, $x \in A = [0, 1]$, $x \in \mathbb{N}$.

38. (a) State and prove the Cauchy criteria for uniform convergence of a sequence of functions.

(b) Prove that $\beta(m, n) = \int_{x=0}^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$.

(1 × 15 = 15 marks)